

S. NITHIESH

Generative AI Engineer | Machine Learning Engineer

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Summary

AI & Machine Learning Engineer skilled in Python, Machine Learning, Deep Learning, NLP, and Generative AI. Experienced in building end-to-end AI solutions including Retrieval-Augmented Generation (RAG) systems, LLM-powered applications, and predictive ML models, with hands-on data-pipeline and deployment experience. Focused on developing production-ready AI systems that solve real-world business problems through automation and applied machine learning.

Experience

Finance Intern

Dec 2024 – Feb 2025

Tinkas Industries, Chennai, TN

- Assisted in the preparation of detailed financial statements and reports, ensuring accuracy and compliance.
- Conducted variance analysis to identify discrepancies and support strategic budgetary planning.
- Collaborated with cross-functional departments to analyze the financial implications of operational decisions.
- Gained deep insights into financial operations, enhancing core analytical and decision-making skills.

Technical Skills

- Programming:** Python, SQL
- Machine Learning:** Scikit-Learn, XGBoost, Gradient Boosting, Random Forest
- Deep Learning:** PyTorch, TensorFlow, Keras
- Generative AI:** OpenAI API, LangChain, RAG, Prompt Engineering
- NLP:** Transformers, Hugging Face, Text Classification, Tokenization
- Vector Databases:** FAISS, ChromaDB, Pinecone
- Data Science:** Pandas, NumPy, SciPy, Statsmodels
- Databases:** MySQL, PostgreSQL
- Tools & Version Control:** Git, GitHub, Jupyter Notebook, Google Colab

Education

Master of Science in Data Science & Analytics

Sep 2025 – Present

M S Ramaiah University of Applied Science, Bangalore

Bachelor of Mathematics

Jun 2022 – 2025

Loyola College, Chennai

- Leading Cadet of Loyola NCC (Navy).

Projects

NapSafe – AI-Powered Driver Fatigue Prediction & Smart Rest Pod Recommendation System Python, Pandas, Scikit-learn, Streamlit, Machine Learning, GIS Analytics

- Engineered a predictive fatigue-risk assessment system utilizing driver behavioral metrics, sleep-quality indicators, and operational workload data to estimate exhaustion levels and generate real-time safety recommendations for commercial drivers.
- Developed a machine learning pipeline incorporating Logistic Regression, feature engineering, and risk-scoring algorithms to classify driver fatigue probability, enabling proactive accident prevention and fleet safety monitoring.
- Designed an interactive analytics dashboard with a location-intelligence framework (traffic density, taxi-stand clustering, accident hotspot mapping) to optimize smart rest pod deployment across high-demand transportation corridors.

AI Hallucination Detection & Uncertainty-Aware Conversational QA System

Python, PyTorch, NumPy,

Scikit-learn, Monte Carlo Dropout, Mahalanobis Distance, NLP

- Developed an uncertainty-aware conversational question-answering framework to identify and mitigate hallucinated responses in Large Language Models using Uncertainty Quantification (UQ).
- Implemented Monte Carlo Dropout-based Bayesian inference to estimate predictive uncertainty through multiple stochastic forward passes and variance analysis, complemented by Mahalanobis Distance for Out-of-Distribution (OOD) detection.
- Built a Selective Generation mechanism that abstains from low-confidence predictions, evaluating confidence calibration and F1-score variations to improve reliability and reduce hallucination risk in conversational AI systems.

Quantitative Risk & Performance Dashboard

Python, Pandas, NumPy, SciPy, Matplotlib

- Engineered a fully automated risk-analysis pipeline processing over 10 years of NIFTY 50 historical data to calculate daily/annualized returns, volatility, and Sharpe ratios across 50 constituent equities.
- Developed a production-ready financial dashboard leveraging Matplotlib GridSpec, featuring risk-return scatter plots, kernel density estimations (KDE), and sector-wise performance benchmarking.

Algorithmic Portfolio Optimizer & Asset Pricing Engine

Python, SciPy, QuantLib, YFinance, Scikit-learn,

Statsmodels

- Built a Modern Portfolio Theory (MPT) optimization tool using SciPy's SLSQP optimizer to dynamically generate custom asset weights, maximizing the Sharpe Ratio and mathematically minimizing portfolio risk.
- Designed a comprehensive quantitative pipeline to execute ARCH/GARCH volatility forecasting, Statsmodels OLS linear regression, and QuantLib Black-Scholes European call option pricing.